GRADE 3

Computational Thinking

National Level Examination

SUBJECT CODE | | | |

Sample paper

Total Questions: 30 **Total Marks: 30** **Time: 45 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **CAREFULLY CHECK YOUR NAME, CLASS, ROLL NUMBER, AND OTHER REQUIRED DETAILS ON THE OMR ANSWER SHEET.**
- **DO NOT FORGET TO SIGN THE OMR SHEET. ALSO WRITE YOUR ROLL NO. ON THE QUESTION PAPER BOOKLET.**
- **USE ONLY BLACK/BLUE BALL PEN AND HB PENCIL TO FILL THE OMR ANSWER SHEET.**
- **ALL QUESTIONS ARE COMPULSORY AND CARRY ONE MARK EACH.**
- **SECTION A: ABSTRACT & SPATIAL THINKING (6 QUESTIONS); SECTION B: PATTERN RECOGNITION (6 QUESTIONS); SECTION C: DECOMPOSITION, NUMBERS & DATA (6 QUESTIONS); SECTION D: ALGORITHMIC THINKING & LOGICAL SEQUENCING (6 QUESTIONS); SECTION E: OLYMPIAD CHALLENGE (6 QUESTIONS).**
- **ONLY OMR ANSWER SHEETS WILL BE EVALUATED.**
- **NO NEGATIVE MARKING FOR INCORRECT ANSWERS.**
- **YOU MAY USE THE SPACE PROVIDED ON THE LAST PAGE FOR ROUGH WORK (IF NEEDED).**
- **DIGITAL DEVICES LIKE CALCULATORS, SMART WATCHES, OR MOBILE PHONES ARE NOT ALLOWED.**
- **AFTER THE EXAM, SUBMIT BOTH THE OMR SHEET AND THE QUESTION PAPER TO THE INVIGILATOR.**

SECTION A – ABSTRACT & SPATIAL THINKING

Q1. Riya is facing North. She makes one quarter-turn to her right. Which direction is she facing now?

A. North
B. East
C. South
D. West

Q2. Which shape is seen when you look straight down at the top of a cylinder?

A. Triangle
B. Rectangle
C. Circle
D. Square

Q3. A square sheet is folded once into half. One hole is punched away from the fold. How many holes will appear when the sheet is opened?

A. 1
B. 2
C. 3
D. 4

Q4. Meena walks 3 steps east and then 2 steps west. Where is she now ?

A.1 step east of the start
B. 1step west of the start
C. At the start
D. 5 step east

Q5. A cube is painted on all its faces if it is placed on a table, how many faces can you see at one time?

A. 4
B. 2
C. 6
D. 3

Q6. Two identical cubes are joined together by one complete face. How many faces are exposed altogether?

A. 8
B. 9
C. 10
D. 12

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SECTION B – PATTERN RECOGNITION

Q7. Which number is even ?

A.13
B. 25
C.38
D. 47

Q8. Find the next number: 1, 3, 6, 10, 15, ___

A. 18
B. 19
C. 20
D. 21

Q9. Which letter comes next? Z, X, V, T, ___

A. S
B. R
C. Q
D. P

Q10. Find the missing number: 20, 18, 15, 11, 6, ___

A. 0
B. 1
C. 2
D. 3

Q11. The pattern repeats: ●, ▲, ●, ▲, ●, ▲ ... Which shape will be at the 15th position?

A. ●
B. ▲
C. ■
D. Cannot tell

Q12. What comes next? 3, 6, 5, 10, 9, 18, ___

A. 16
B. 17
C. 18
D. 19

Sample paper

SECTION C – DECOMPOSITION, NUMBERS & DATA

Q13. I am a 3-digit number. My hundreds digit is 3. My tens digit is 2 more than my ones digit. The sum of my digits is 11. What number am I?

A. 335
B. 353
C. 533
D. 355

Q14. A cube is rolled one step to the right ,which part of the cube touches the ground next?

A. top face
B. right face
C. front face
D. back face

Q15. Neha has 18 stickers. She gives 5 stickers to her friend and then receives 3 new stickers. How many stickers does she have now?

A. 13
B. 15
C. 16
D. 21

Q16. Ravi has ₹50. He buys a pencil for ₹18 and an eraser for ₹12. How much money is left?

A. ₹18
B. ₹20
C. ₹22
D. ₹30

Q17. A class chart shows: Red - 4, Blue - 7, Green - 5, Yellow - 6. How many students chose Blue and Yellow altogether?

A. 11
B. 12
C. 13
D. 14

Q18. There are 14 boys and 16 girls in a class. How many students are there altogether?

A. 28
B. 29
C. 30
D. 32

Sample paper

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING

Q19. A number machine follows two steps: Add 3, then multiply by 2. What is the output for input 5?

A. 11
B. 13
C. 16
D. 19

Q20. A machine multiplies a number by 2 and then adds 1. The output is 15. What was the input?

A. 6
B. 7
C. 8
D. 9

Q21. Start with 3. Repeat these two steps twice: Multiply by 2, then subtract 1. What is the final number?

A. 7
B. 8
C. 9
D. 10

Q22. A robot moves 2 squares North, turns right and moves 3 squares. Where is it from its starting point?

A. North-East
B. North-West
C. South-East
D. South-West

Q23. A robot starts facing East. It turns left, moves 2 steps, turns right and moves 1 step. Where is it from its starting point?

A. North-East
B. North-West
C. South-East
D. South-West

Q24. Four actions must happen in this order: Wash hands before opening the lunchbox; open the lunchbox before eating; close the lunchbox after eating. Which action is second?

A. Eat lunch
B. Open the lunchbox
C. Close the lunchbox
D. Wash hands

SECTION E – OLYMPIAD CHALLENGE

Q25. I am a 3-digit number with different digits. My hundreds digit is 5. My tens digit is 1 more than my ones digit. The sum of my digits is 12. What number am I?

A. 534
B. 543
C. 552
D. 453

Q26. A robot starts facing North. It turns right and moves 2 steps, turns left and moves 3 steps, then turns left and moves 2 steps. Where is it from the starting point?

A. 3 steps North
B. 3 steps South
C. 2 steps East
D. 2 steps West

Q27. Find the next number: 2, 4, 8, 7, 14, 28, 27, ___

A. 29
B. 54
C. 56
D. 81

Q28. Four children P, Q, R and S are of different ages. P is older than Q. R is younger than Q. S is older than P. Who is the second oldest?

A. P
B. Q
C. R
D. S

Q29. Two stars together have a value of 12. One triangle has a value 2 less than one star. What is the value of one star plus one triangle?

A. 8
B. 9
C. 10
D. 12

Q30. Use this rule: If the number is odd, multiply it by 2. If the number is even, add 4. Start with 5 and apply the rule three times. What is the final number?

A. 14
B. 16
C. 18
D. 20

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